Exercise 9.10: The Historical Hölder Inequality

How weighted Jensen and modern Hölder are the same idea in disguise

A step-by-step tutorial for a high-school reader

What you will learn

By the end of this tutorial, you will be able to:

- understand weights and weighted averages;
- recognize why the function $t \mapsto t^p$ is convex for $p > 1$;
- apply weighted Jensen's inequality to prove Hölder's historical formula;
- understand conjugate exponents p, q satisfying $1/p + 1/q = 1$;
- turn the modern Hölder inequality into the historical one by splitting a factor;
- reverse that substitution and recover the modern inequality;
- identify the equality condition in both forms.

Contents

1 The exercise, stated carefully

The problem

Let $w_1, \dots, w_n \ge 0$, let $y_1, \dots, y_n \ge 0$, and let $p > 1$. Prove from weighted Jensen's inequality that

$$\boxed{\sum_{k=1}^{n} w_k y_k \le \left(\sum_{k=1}^{n} w_k\right)^{(p-1)/p} \left(\sum_{k=1}^{n} w_k y_k^p\right)^{1/p}.} \tag{$\mathrm{H}_{\mathrm{hist}}$}$$

Then show that this historical form is equivalent to the modern Hölder inequality. Namely, if $p, q > 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1,$$

show that ($\mathrm{H}_{\mathrm{hist}}$) and

$$\boxed{\sum_{k=1}^{n} a_k b_k \le \left(\sum_{k=1}^{n} a_k^p\right)^{1/p} \left(\sum_{k=1}^{n} b_k^q\right)^{1/q}} \tag{$\mathrm{H}_{\mathrm{modern}}$}$$

imply one another for nonnegative sequences (a_k) and (b_k).

What if the entries can be negative?

The exercise assumes nonnegative numbers because real powers such as a_k^p may be undefined when $a_k < 0$ and p is not an integer. For arbitrary real sequences, the standard form is

$$\left|\sum_{k=1}^{n} a_k b_k\right| \le \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |b_k|^q\right)^{1/q}.$$

It follows from the nonnegative version by applying it to $|a_k|$ and $|b_k|$, then using

$$\left|\sum a_k b_k\right| \le \sum |a_k b_k|.$$

2 Background: the ideas used in the proof

2.1 Weights and weighted averages

Suppose the numbers $y_1, \dots, y_n$ are measurements and the nonnegative numbers $w_1, \dots, w_n$ tell us how important each measurement is. Their total weight is

$$W := \sum_{k=1}^{n} w_k. \tag{1}$$

When $W > 0$, the weighted average is

$$\overline{y}_w := \frac{w_1 y_1 + \cdots + w_n y_n}{w_1 + \cdots + w_n} = \frac{\sum_{k=1}^{n} w_k y_k}{W}. \tag{2}$$

For example, let $y_1 = 2$, $y_2 = 4$, with weights $w_1 = 1$, $w_2 = 3$. Then

$$\overline{y}_w = \frac{1 \cdot 2 + 3 \cdot 4}{1 + 3} = \frac{14}{4} = 3.5.$$

The answer is closer to 4 than to 2 because the value 4 has three times as much weight.

It is often convenient to normalize the weights:

$$\lambda_k := \frac{w_k}{W}. \tag{3}$$

Then

$$\lambda_k \geq 0, \qquad \sum_{k=1}^{n} \lambda_k = 1, \qquad \overline{y}_w = \sum_{k=1}^{n} \lambda_k y_k. \tag{4}$$

Thus the λ_k's are percentages written as decimals: they describe how the total weight is divided among the entries.

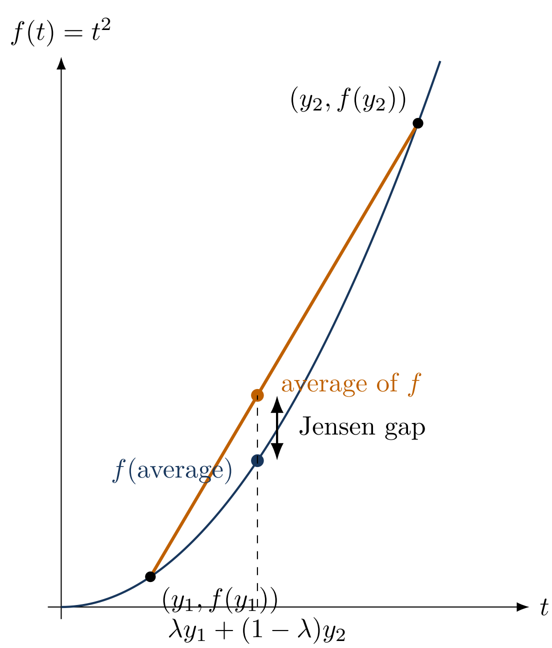
2.2 Convex functions

A convex function bends upward like a bowl. The important geometric fact is that the straight chord joining two points of its graph lies on or above the graph.

For this problem we use

$$f(t) = t^p, \qquad t \geq 0, \qquad p > 1. \tag{5}$$

This function is convex. For example, when $p = 2$, its graph is the parabola $f(t) = t^2$.



The lower point is f evaluated at a weighted average of the inputs. The upper point is the same weighted average of the output heights. Convexity says the lower point cannot rise above the upper one.

Optional calculus check

If you know derivatives, then for $t > 0$,

$$f''(t) = p(p-1)t^{p-2} \geq 0.$$

A nonnegative second derivative means the graph bends upward. Continuity at $t = 0$ extends convexity to all of $[0, \infty)$.

2.3 Weighted Jensen's inequality

Weighted Jensen inequality

Let f be convex, and let $\lambda_k \geq 0$ satisfy $\sum_{k=1}^{n} \lambda_k = 1$. Then

$$\boxed{f\left(\sum_{k=1}^{n} \lambda_k x_k\right) \leq \sum_{k=1}^{n} \lambda_k f(x_k).} \tag{J}$$

In words:

$$\boxed{\text{function of the weighted average} \;\leq\; \text{weighted average of the function values.}}$$

For our choice $f(t) = t^p$, Jensen becomes

$$\left(\sum_{k=1}^{n} \lambda_k y_k\right)^p \leq \sum_{k=1}^{n} \lambda_k y_k^p. \tag{6}$$

This single inequality is the engine of the first proof.

2.4 Conjugate exponents

Conjugate exponents

Numbers $p, q > 1$ are called conjugate exponents when

$$\frac{1}{p} + \frac{1}{q} = 1. \tag{7}$$

If $p > 1$ is given, its conjugate is

$$q = \frac{p}{p-1}. \tag{8}$$

Some examples are

p	q	$1/p + 1/q$
2	2	$1/2 + 1/2 = 1$
3	$3/2$	$1/3 + 2/3 = 1$
4	$4/3$	$1/4 + 3/4 = 1$

Two identities will be used repeatedly:

$$\frac{1}{q} = 1 - \frac{1}{p} = \frac{p-1}{p}, \tag{9}$$

and

$$\frac{q}{p} = q - 1 = \frac{1}{p-1}. \tag{10}$$

For instance, the first identity explains why the historical exponent $(p-1)/p$ is really $1/q$.

3 First goal: derive the historical inequality from Jensen

We now prove

$$\sum_{k=1}^{n} w_k y_k \le \left(\sum_{k=1}^{n} w_k\right)^{(p-1)/p} \left(\sum_{k=1}^{n} w_k y_k^p\right)^{1/p}. \tag{11}$$

Step 0: handle the all-zero weight case

Let

$$W = \sum_{k=1}^{n} w_k.$$

If $W = 0$, then every $w_k = 0$, because the weights are nonnegative. Both sides of (11) are then 0, so the inequality is true.
From now on, assume $W > 0$.

Step 1: turn the weights into normalized weights

Define

$$\lambda_k := \frac{w_k}{W}. \tag{12}$$

Then $\lambda_k \ge 0$ and

$$\sum_{k=1}^{n} \lambda_k = \frac{\sum_{k=1}^{n} w_k}{W} = \frac{W}{W} = 1. \tag{13}$$

Therefore the λ_k's satisfy the hypotheses of weighted Jensen.

Step 2: apply Jensen to $f(t) = t^p$

Since $p > 1$, the function $f(t) = t^p$ is convex on $[0, \infty)$. Jensen gives

$$\left(\sum_{k=1}^{n} \lambda_k y_k\right)^p \le \sum_{k=1}^{n} \lambda_k y_k^p. \tag{14}$$

Substitute $\lambda_k = w_k/W$:

$$\left(\frac{\sum_{k=1}^{n} w_k y_k}{W}\right)^p \le \frac{\sum_{k=1}^{n} w_k y_k^p}{W}. \tag{15}$$

Step 3: take the positive p-th root

All quantities in (15) are nonnegative. The function $t \mapsto t^{1/p}$ is increasing on $[0, \infty)$, so taking the p-th root preserves the inequality:

$$\frac{\sum_{k=1}^{n} w_k y_k}{W} \le \left(\frac{\sum_{k=1}^{n} w_k y_k^p}{W}\right)^{1/p}. \tag{16}$$

Step 4: multiply by W and simplify

Multiplying (16) by $W > 0$ gives

$$\sum_{k=1}^{n} w_k y_k \le W \left(\frac{\sum_{k=1}^{n} w_k y_k^p}{W} \right)^{1/p}. \tag{17}$$

Now separate the power of W:

$$\begin{aligned} W \left(\frac{\sum w_k y_k^p}{W} \right)^{1/p} &= W \cdot W^{-1/p} \left(\sum w_k y_k^p \right)^{1/p} \\ &= W^{1-1/p} \left(\sum w_k y_k^p \right)^{1/p} \\ &= W^{(p-1)/p} \left(\sum w_k y_k^p \right)^{1/p}. \end{aligned} \tag{18}$$

Because $W = \sum w_k$, this is exactly

$$\boxed{\sum_{k=1}^{n} w_k y_k \le \left(\sum_{k=1}^{n} w_k \right)^{(p-1)/p} \left(\sum_{k=1}^{n} w_k y_k^p \right)^{1/p}.}$$

The proof in one line

For $W = \sum w_k > 0$, weighted Jensen says

$$\left(\frac{\sum w_k y_k}{W} \right)^p \le \frac{\sum w_k y_k^p}{W}.$$

Taking the p-th root and multiplying by W gives the historical Hölder inequality.

3.1 A numerical example

Take

$$p = 2, \qquad (w_1, w_2) = (1, 3), \qquad (y_1, y_2) = (2, 4).$$

The left side is

$$\sum w_k y_k = 1 \cdot 2 + 3 \cdot 4 = 14.$$

The right side is

$$(1+3)^{1/2} \left(1 \cdot 2^2 + 3 \cdot 4^2 \right)^{1/2} = 2\sqrt{52} \approx 14.422.$$

Thus

$$14 \le 14.422.$$

In normalized Jensen form, the same calculation is

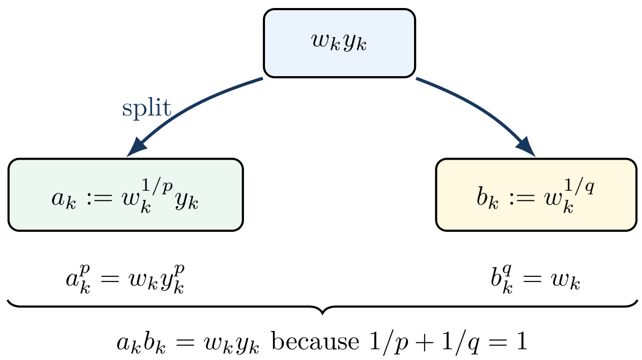
$$\left(\frac{14}{4} \right)^2 = 3.5^2 = 12.25 \le \frac{52}{4} = 13.$$

The historical inequality is therefore just this Jensen comparison rewritten without fractions.

4 The factor-splitting picture

The equivalence with modern Hölder becomes much easier when we notice the following exact factorization. Let q be conjugate to p. Since $1/p + 1/q = 1$,

$$w_k y_k = w_k^{1/p} w_k^{1/q} y_k = \left(w_k^{1/p} y_k\right)\left(w_k^{1/q}\right). \tag{19}$$



This diagram already contains the proof that modern Hölder implies the historical version: apply modern Hölder to the two pieces.

5 Modern Hölder implies the historical form

Assume the modern Hölder inequality is known. Let $p > 1$, and define its conjugate exponent

$$q := \frac{p}{p-1}. \tag{20}$$

Then $1/p + 1/q = 1$ and $1/q = (p-1)/p$.

Step 1: split each term $w_k y_k$

Define two new sequences

$$a_k := w_k^{1/p} y_k, \qquad b_k := w_k^{1/q}. \tag{21}$$

Then

$$a_k b_k = w_k^{1/p+1/q} y_k = w_k y_k. \tag{22}$$

Also,

$$a_k^p = w_k y_k^p, \qquad b_k^q = w_k. \tag{23}$$

Step 2: apply modern Hölder

Modern Hölder gives

$$\begin{aligned}\sum_{k=1}^{n} w_k y_k &= \sum_{k=1}^{n} a_k b_k \\ &\le \left(\sum_{k=1}^{n} a_k^p\right)^{1/p} \left(\sum_{k=1}^{n} b_k^q\right)^{1/q} \\ &= \left(\sum_{k=1}^{n} w_k y_k^p\right)^{1/p} \left(\sum_{k=1}^{n} w_k\right)^{1/q}.\end{aligned} \tag{24}$$

Finally, use $1/q = (p-1)/p$:

$$\sum_{k=1}^{n} w_k y_k \le \left(\sum_{k=1}^{n} w_k\right)^{(p-1)/p} \left(\sum_{k=1}^{n} w_k y_k^p\right)^{1/p}.$$

This is the historical inequality.

Why this direction is especially simple

The weights are divided between the two modern Hölder factors:

$$w_k = w_k^{1/p} w_k^{1/q}.$$

The y_k is placed entirely in the p-factor. Raising the first factor to the p-th power produces $w_k y_k^p$, while raising the second to the q-th power produces w_k.

6 The historical form implies modern Hölder

Now assume the historical inequality is known for every set of nonnegative weights and values. Let $a_k, b_k \ge 0$, and let $p, q > 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1. \tag{25}$$

We want to prove

$$\sum_{k=1}^{n} a_k b_k \le \left(\sum_{k=1}^{n} a_k^p\right)^{1/p} \left(\sum_{k=1}^{n} b_k^q\right)^{1/q}. \tag{26}$$

The idea is to reverse the factor split from the previous section. We want

$$a_k = w_k^{1/p} y_k, \qquad b_k = w_k^{1/q}. \tag{27}$$

The second equation suggests

$$w_k = b_k^q. \tag{28}$$

Then the first equation suggests

$$y_k = \frac{a_k}{w_k^{1/p}} = \frac{a_k}{b_k^{q/p}}. \tag{29}$$

What about an index with $b_k = 0$?

Formula (29) would divide by zero. Such indices cause no real difficulty because $a_k b_k = 0$, so they contribute nothing to the left side. We may apply the argument only to the set

$$I := \{k : b_k > 0\}.$$

Afterward, enlarging $\sum_{k\in I} a_k^p$ to $\sum_{k=1}^n a_k^p$ can only make the right side larger. If every $b_k = 0$, the desired inequality is simply $0 \le 0$.

Assume therefore that I is nonempty, and for $k \in I$ define

$$w_k := b_k^q, \qquad y_k := \frac{a_k}{b_k^{q/p}}. \tag{30}$$

Step 1: simplify the left side of the historical inequality

For $k \in I$,

$$\begin{aligned} w_k y_k &= b_k^q \frac{a_k}{b_k^{q/p}} \\ &= a_k b_k^{q-q/p}. \end{aligned}$$

Now

$$q - \frac{q}{p} = q\left(1 - \frac{1}{p}\right) = q\left(\frac{1}{q}\right) = 1. \tag{31}$$

Therefore

$$w_k y_k = a_k b_k. \tag{32}$$

Step 2: simplify the powered sum

Again for $k \in I$,

$$\begin{aligned} w_k y_k^p &= b_k^q \left(\frac{a_k}{b_k^{q/p}}\right)^p \\ &= b_k^q \frac{a_k^p}{b_k^q} \\ &= a_k^p. \end{aligned} \tag{33}$$

Also,

$$\sum_{k\in I} w_k = \sum_{k\in I} b_k^q = \sum_{k=1}^n b_k^q, \tag{34}$$

because $b_k^q = 0$ outside I.

Step 3: apply the historical inequality

Applying (H_{hist}) over I gives

$$\begin{aligned}\sum_{k=1}^{n} a_k b_k &= \sum_{k\in I} w_k y_k \\ &\le \left(\sum_{k\in I} w_k\right)^{(p-1)/p} \left(\sum_{k\in I} w_k y_k^p\right)^{1/p} \\ &= \left(\sum_{k=1}^{n} b_k^q\right)^{(p-1)/p} \left(\sum_{k\in I} a_k^p\right)^{1/p} \\ &\le \left(\sum_{k=1}^{n} b_k^q\right)^{1/q} \left(\sum_{k=1}^{n} a_k^p\right)^{1/p}.\end{aligned} \tag{35}$$

In the last line, we used $(p-1)/p = 1/q$. Reordering the two factors gives the modern Hölder inequality.

The reverse substitution

To recover modern Hölder from the historical form, solve

$$a_k = w_k^{1/p} y_k, \qquad b_k = w_k^{1/q}.$$

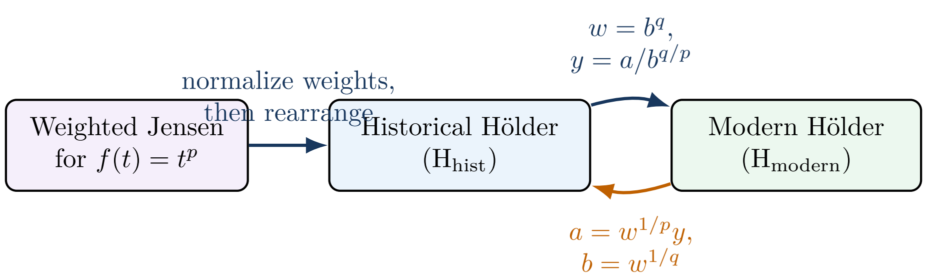
This leads to

$$\boxed{w_k = b_k^q, \qquad y_k = \frac{a_k}{b_k^{q/p}}.}$$

That substitution is the main creative step. Everything afterward is exponent simplification.

7 The complete equivalence loop

The three inequalities fit together as follows.



Thus historical and modern Hölder are not merely similar. They are the same inequality written using different variables.

8 Equality: when do the two sides match?

The function t^p is strictly convex for $p > 1$. Therefore equality in weighted Jensen occurs precisely when all values y_k carrying positive weight are equal:

$$w_k > 0 \quad \Longrightarrow \quad y_k = c \tag{36}$$

for some constant $c \geq 0$. This is also the equality condition in the historical inequality.

To translate it into the modern variables, use

$$w_k = b_k^q, \qquad y_k = \frac{a_k}{b_k^{q/p}}.$$

Equality requires

$$\frac{a_k}{b_k^{q/p}} = c \qquad \text{whenever } b_k > 0. \tag{37}$$

Therefore

$$a_k = c\, b_k^{q/p}.$$

Raising both sides to the p-th power gives

$$\boxed{a_k^p = c^p b_k^q.} \tag{38}$$

So, apart from zero entries, equality in modern Hölder occurs when the powered sequences (a_k^p) and (b_k^q) are proportional.

8.1 An equality example

Take $p = q = 2$,

$$a = (1, 2, 3), \qquad b = (2, 4, 6) = 2a.$$

Then $a_k^2 = (1/4)b_k^2$, so the proportionality condition holds. Indeed,

$$\sum a_k b_k = 2(1^2 + 2^2 + 3^2) = 28,$$

and

$$\left(\sum a_k^2\right)^{1/2} \left(\sum b_k^2\right)^{1/2} = \sqrt{14}\sqrt{56} = \sqrt{784} = 28.$$

9 A second numerical example: seeing the two forms agree

Let $p = q = 2$, and choose

$$w = (1, 4), \qquad y = (3, 2).$$

The factor split gives

$$a_k = w_k^{1/2} y_k, \qquad b_k = w_k^{1/2}.$$

Thus

$$a = (3, 4), \qquad b = (1, 2).$$

The historical left side is

$$\sum w_k y_k = 1 \cdot 3 + 4 \cdot 2 = 11,$$

while the modern left side is the same number:

$$\sum a_k b_k = 3 \cdot 1 + 4 \cdot 2 = 11.$$

The historical right side is

$$(1+4)^{1/2}(1 \cdot 3^2 + 4 \cdot 2^2)^{1/2} = \sqrt{5}\sqrt{25} = 5\sqrt{5}.$$

The modern right side is also

$$\sqrt{3^2+4^2}\sqrt{1^2+2^2} = 5\sqrt{5}.$$

The two formulas give literally the same numerical inequality:

$$11 \le 5\sqrt{5}.$$

10 Common mistakes and how to avoid them

Mistake 1: forgetting the case $\sum w_k = 0$

Normalized weights w_k/W make sense only when $W > 0$. Check $W = 0$ first. Because the weights are nonnegative, $W = 0$ means every weight is zero, and the result is immediate.

Mistake 2: applying Jensen to the wrong function

The needed choice is

$$f(t) = t^p.$$

Jensen then produces y_k^p, exactly the power appearing in the historical inequality. Choosing $f(t) = t^{1/p}$ would use a concave function and reverse the direction.

Mistake 3: losing the factor $W^{(p-1)/p}$

From

$$\frac{\sum w_k y_k}{W} \le \left(\frac{\sum w_k y_k^p}{W}\right)^{1/p},$$

multiplying by W gives

$$W \cdot W^{-1/p} = W^{1-1/p} = W^{(p-1)/p},$$

not simply W.

Mistake 4: using $w_k = b_k^p$ in the reverse direction

To make the second modern norm equal $(\sum b_k^q)^{1/q}$, the correct choice is

$$w_k = b_k^q,$$

not b_k^p. The exponent q is forced by the equation $b_k = w_k^{1/q}$.

Mistake 5: dividing by b_k when $b_k = 0$

When reversing the substitution, restrict first to indices with $b_k > 0$. Terms with $b_k = 0$ contribute $a_k b_k = 0$ to the left side and can safely be omitted.

11 A compact solution suitable for homework

Concise proof

Let $W = \sum_{k=1}^n w_k$. If $W = 0$, the historical inequality is trivial. Assume $W > 0$, and set $\lambda_k = w_k/W$. Since $t \mapsto t^p$ is convex for $p > 1$, weighted Jensen gives

$$\left(\sum_{k=1}^n \lambda_k y_k\right)^p \le \sum_{k=1}^n \lambda_k y_k^p.$$

Hence

$$\left(\frac{\sum w_k y_k}{W}\right)^p \le \frac{\sum w_k y_k^p}{W}.$$

Taking the p-th root and multiplying by W,

$$\sum w_k y_k \le W^{1-1/p}\left(\sum w_k y_k^p\right)^{1/p} = W^{(p-1)/p}\left(\sum w_k y_k^p\right)^{1/p}.$$

This proves the historical form.
Now let $q = p/(p-1)$. If the modern inequality is assumed, apply it to

$$a_k = w_k^{1/p} y_k, \qquad b_k = w_k^{1/q}.$$

Then $a_k b_k = w_k y_k$, $a_k^p = w_k y_k^p$, and $b_k^q = w_k$, so modern Hölder becomes the historical inequality. Conversely, assume the historical inequality. Given nonnegative a_k, b_k, omit indices with $b_k = 0$, and put

$$w_k = b_k^q, \qquad y_k = \frac{a_k}{b_k^{q/p}}.$$

Then

$$w_k y_k = a_k b_k, \qquad w_k y_k^p = a_k^p, \qquad \sum w_k = \sum b_k^q.$$

The historical inequality therefore gives

$$\sum a_k b_k \le \left(\sum b_k^q\right)^{(p-1)/p}\left(\sum a_k^p\right)^{1/p} = \left(\sum a_k^p\right)^{1/p}\left(\sum b_k^q\right)^{1/q},$$

which is the modern Hölder inequality. Thus the two forms are equivalent.

12 Self-check questions

Try these before reading the answers.

1. If $p = 3$, what is its conjugate exponent q?
2. For $p = 4$, simplify $(p-1)/p$, and compare it with $1/q$.

3. Why does $\sum w_k = 0$ force every $w_k = 0$?
4. Starting from Jensen,
$$\left(\frac{\sum w_k y_k}{W}\right)^p \le \frac{\sum w_k y_k^p}{W},$$
explain where the power $W^{(p-1)/p}$ comes from.
5. Verify that with
$$a_k = w_k^{1/p} y_k, \qquad b_k = w_k^{1/q},$$
one has $a_k^p = w_k y_k^p$, $b_k^q = w_k$, and $a_k b_k = w_k y_k$.
6. In the reverse substitution, why is $w_k = b_k^q$ a natural choice?
7. What is the equality condition in modern Hölder for positive entries?

Answers

1. $q = p/(p-1) = 3/2$.
2. $(p-1)/p = 3/4$. The conjugate of 4 is $q = 4/3$, so $1/q = 3/4$.
3. A sum of nonnegative numbers can equal zero only if every term is zero.
4. Taking the p-th root produces a factor $W^{-1/p}$ on the right. Multiplying both sides by W gives $W \cdot W^{-1/p} = W^{1-1/p} = W^{(p-1)/p}$.
5. Raise each definition to the matching power and use $1/p + 1/q = 1$.
6. We want $b_k = w_k^{1/q}$. Raising both sides to the q-th power forces $w_k = b_k^q$.
7. There is a constant $C \ge 0$ such that $a_k^p = C b_k^q$ for all relevant indices.

13 Final summary

Three formulas to remember

Weighted Jensen with $f(t) = t^p$ gives
$$\left(\frac{\sum w_k y_k}{\sum w_k}\right)^p \le \frac{\sum w_k y_k^p}{\sum w_k} \quad \Longrightarrow \quad \sum w_k y_k \le \left(\sum w_k\right)^{(p-1)/p} \left(\sum w_k y_k^p\right)^{1/p}.$$

For conjugate p, q, the two changes of variables are
$$\boxed{a_k = w_k^{1/p} y_k, \quad b_k = w_k^{1/q}} \qquad \text{and} \qquad \boxed{w_k = b_k^q, \quad y_k = \frac{a_k}{b_k^{q/p}}} \quad (b_k > 0).$$

They convert the historical and modern forms into one another.